

ROMA TRE · A.Y. 2026/27

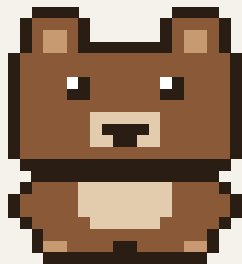
Policy Evaluation Lab

Lecture 1: Causality (I)

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PART 1

Introduction



Two types of empirical research

We can distinguish two types of empirical research in economics:

1. **descriptive analysis**
2. **causal inference**

According to Angrist (Nobel Prize 2021) and Krueger (1999), descriptive analysis:

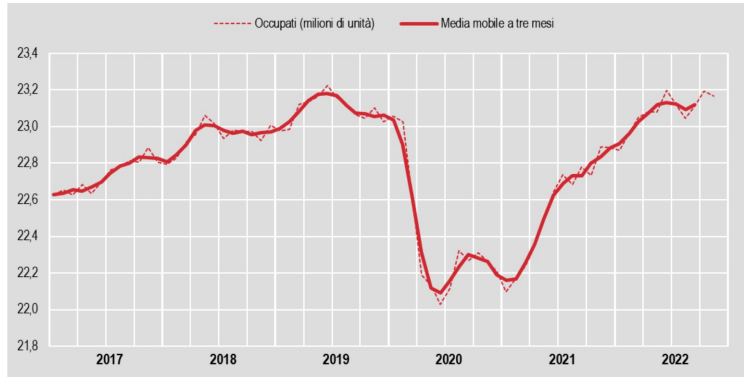
- can establish facts about the labour market that need to be explained by theoretical reasoning, and yield new insights into economic trends
- includes work on trends in poverty rates, labour force participation and wage levels



Descriptive analysis: example 1

FIGURA 1. OCCUPATI

Gennaio 2017 – novembre 2022, valori assoluti in milioni, dati destagionalizzati



Descriptive analysis: example 2

Descriptive statistics

Variable	Mean	S.D.	Min.	Max.	Variable	Mean	S.D.	Min.	Max.
female	0.706	0.456	0	1	mother HE degree	0.228	0.420	0	1
age	20.815	2.671	19	42	father working	0.934	0.247	0	1
residence	0.211	0.409	0	1	father HE degree	0.217	0.413	0	1
lyceum	0.701	0.459	0	1	smoke	0.315	0.465	0	1
scholarship eligibility	0.244	0.431	0	1	family illness	0.581	0.494	0	1
HS grade	82.788	11.644	60	100	listening music	0.451	0.498	0	1
grade	20.896	4.941	8	31	going to concert	0.576	0.495	0	1
cramming	0.532	0.500	0	1	using social network	0.456	0.499	0	1
mother working	0.494	0.501	0	1					

S.D.: standard deviation. Full sample of 184 HE students in the 1st year of Law.



Causal inference

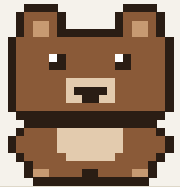
Causal inference:

- seeks to determine the effects of particular interventions or policies, or to estimate features of the behavioural relationships suggested by economic theory
- Causal inference and descriptive analysis are not competing methods: they are often **complementary**
- The “experimentalist” approach, also called quasi-experimental, uses economic theory to frame causal questions
- Angrist and Krueger (Handbook, 1999):
 - “...the ‘experimentalist’ approach, ... puts front and center the problem of identifying causal effects from specific events or situations.”



PART 2

Experimentalist approach



Experimentalist approach (I)

Consider the following questions:

- **Medical therapies:** how can we establish whether a drug is effective?
- **College education:** does college increase your future earnings? Does college add anything to your innate abilities? Can we estimate the returns to schooling in general? For which group in the population?
- **Labour market programmes:** does training increase the employment probability of jobless workers? Do work incentives for single mothers with children increase labour force participation?



Experimentalist approach (II)

- **Army:** does military service increase or reduce earnings and employment probabilities? Health? Life expectancy?
- **Educational policies:** can the offer of student loans increase college education and earnings of poor high school graduates?
- **Public policies:**
 - effect of changes in electoral rules on public spending and taxation
 - effect of joining a union of municipalities on local expenditure
 - effect of fiscal centralisation on local per capita current expenditure and revenue

The answers to these questions involve the identification and measurement of **causal links**



Causation: an old problem in philosophy and statistics

- **Adam Smith** wrote about the causes of the wealth of nations
- **David Hume** described causation as a sequence of temporal events in which, had the first event not occurred, subsequent ones would not have either
- **J.S. Mill** wrote about the method of difference, which is similar to the idea of causation as a comparison among counterfactuals



Do not confuse correlation with causality

- When the rooster crows, the sun soon after rises
- We know the rooster didn't cause the sun to rise
- If the rooster had been eaten by the farmer's cat, the sun would still have risen

Does correlation imply causation? In general, **no**.

- Our interest is to understand why correlations, particularly in observational data, are unlikely to represent a causal relationship

Correlation is not causation



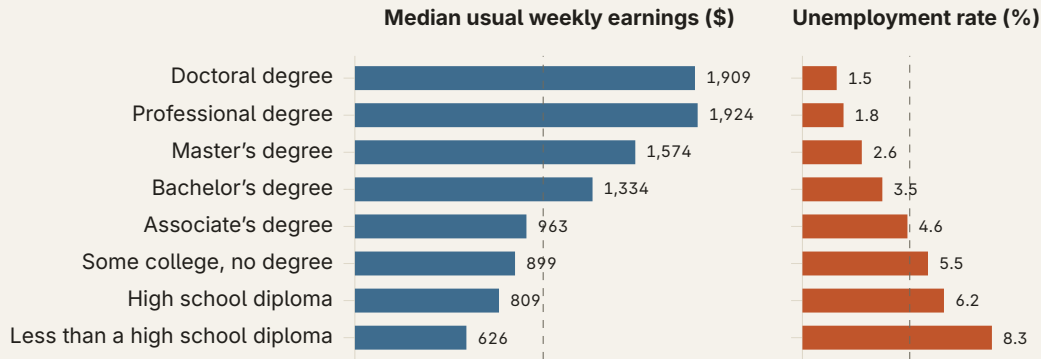
Spurious correlations

- Two variables can move together without one causing the other
- Let's look at some examples together:
 - **Spurious correlations:** <https://www.tylervigen.com/spurious-correlations>



Example 3a: education and earnings (US)

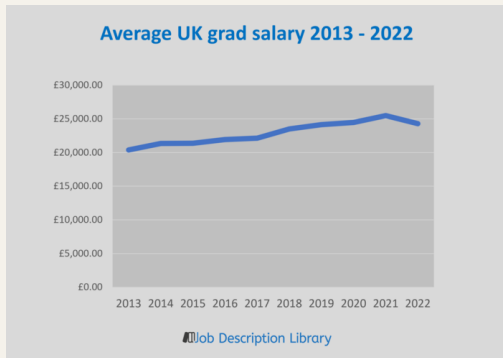
Earnings and unemployment rates by educational attainment, 2021



Note: data are for persons aged 25 and over. Earnings are for full-time wage and salary workers. Source: U.S. Bureau of Labor Statistics, Current Population Survey.



Example 3b: graduates vs non-graduates (UK, 2021)



- The Department for Education shows that the **median** salary for graduates over their **entire** working career is £34,000, and £25,000 for non-graduates
 - Graduates earn on average £9,000 per year more than non-graduates
- lifetime earnings gap of £321,000



Example: returns to higher education

- Returns to higher education (HE): HE degree vs high school (HS) diploma
- £321,000 over a lifetime (UK) and 61% ↑ (US, 2021)
- Is this the **causal effect** of higher education on wages?
- No, it is a strong **correlation**: other factors affect education (e.g. ability, effort) and bias its effect on wages
- Finding a causal effect requires controlling for, and trying to remove, all those **confounding factors**



PART 3

Potential outcomes



States of the world

A causal effect is the comparison between two **states of the world**:

1. In the first state (the “**actual**” state of the world), an individual takes an HE degree and we observe their (average) wage
2. In the second state (the “**counterfactual**” state of the world), that **same** person does not take an HE degree, and we observe their HS (average) wage
 - The causal effect of HE is the difference in wages between the two states of the world: $\text{Wage(HE)} - \text{Wage(HS)}$ for the **same** individual
 - Since the individual is the same, there are no confounding factors (e.g. same ability, same effort in both states)



Causal effects

To wonder how life would be different if one single event had been different is to think in terms of **counterfactuals**.

- Counterfactuals are **not** realised in history, because they are hypothetical states of the world
- History is a sequence of observable events, hence we are missing data on the counterfactual outcome
- Potential outcomes are a set of *ex ante* possibilities, but once a decision is made, all but one outcome disappear



Causal framework: treatment

Let D_i be a **treatment indicator** (e.g. HE degree vs HS diploma), where:

- $D_i = 1$ if individual i receives the treatment (e.g. gets an HE degree)
 - i belongs to the **treated group**
- $D_i = 0$ if individual i does not (e.g. has an HS diploma)
 - i belongs to the **control group**



Potential outcomes

For individual i we postulate the existence of two **potential outcomes**:

- $Y_i(1)$ denotes the outcome (earnings level) that would be realised by individual i if they get an HE degree, $D_i = 1$
- $Y_i(0)$ denotes the outcome (earnings level) that would be realised by individual i if they do not get an HE degree, $D_i = 0$



Individual-level causal effect

We are interested in the difference

$$\tau_i = Y_i(1) - Y_i(0)$$

- which is the **causal effect**, or treatment effect, of HE on the earnings of individual i
- This is what we would measure if we could go back in time and push i onto the road not taken



Counterfactuals

- However, individual i can either participate or not participate in the programme, but not both
- Only one of the two potential outcomes can be realised: the **actual outcome** Y_i
- If i participates [does not participate] in the programme, $Y_i = Y_i(1)$ [$Y_i = Y_i(0)$] will be realised
- and $Y_i(0)$ [$Y_i(1)$] will be the unobserved outcome, called the **counterfactual**



Switching equation

- The observable or “actual” outcome Y_i is distinct from the potential outcomes,
- because it is realised: actual, historical, empirical

A unit's observable outcome Y_i is a function of its potential outcomes, determined according to the **switching equation**:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

If $D_i = 1 \longrightarrow Y_i = Y_i(1)$, and if $D_i = 0 \longrightarrow Y_i = Y_i(0)$



Fundamental problem of causal inference

It is impossible to observe for the same individual i the values $D_i = 1$ and $D_i = 0$, as well as the values $Y_i(1)$ and $Y_i(0)$; therefore, it is impossible to observe the effect of D on Y for individual i (Holland, 1986).

We cannot infer the effect of the treatment because we do not have the **counterfactual** evidence, i.e. what would have happened in the absence of treatment



Does the use of averages help?

- The causal effect for a single individual i cannot be observed
- The statistical solution proposes methods to compute the **average causal effect** for the entire population or for some interesting subgroups

Define:

- **treated group**: the group that participates in the programme
- **control group**: the group unaffected by the programme



Average treatment effect (ATE)

- This is the most popular treatment effect:
- the population expectation of the unit-level causal effect $Y_i(1) - Y_i(0)$
- In other words, suppose you pick a person at random in the population and expose them to training
- The expected effect on employment for this person is:

$$ATE = E[\tau_i] = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)]$$

As with individual-level treatment effects, the ATE requires both potential outcomes for each unit i . Thus, the ATE cannot be **calculated**, but it can be **estimated**.



Average treatment effect on the treated (ATT)

- Another popular parameter is the average over the subpopulation of those who are actually treated
- In practice, it is the difference between the average outcome under treatment, which we observe (employment of trainees),
- minus the average outcome in the counterfactual situation of no treatment, which we do not observe (employment of trainees had they not been trained)

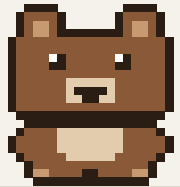
$$ATT = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1]$$

The ATT is also unknowable, because it requires two observations per treated unit i .



PART 4

Application: job training



Policy application: labour market programmes

Does training increase the employment probability of jobless workers?

- Government training programmes usually expose disadvantaged workers (long-term unemployed, drug addicts, ex-offenders, low-income workers) to training
- The idea is to increase their employment and earnings
- In the US, \$4 billion of federal spending per year goes to job training programmes

What is the **causal effect** of on-the-job training (OJT) on the earnings of disadvantaged workers?



Example: OJT

- A training programme to ease the transition of disadvantaged workers into the labour market starts in 1979
- Effect of OJT on annual earnings in 1979, for individuals observed in 1975, before training

	Treatment group	Control group
Annual pre-training earnings (1975)	$\bar{Y}_B(1) = 1512$	$\bar{Y}_B(0) = 1481$
Annual post-training earnings (1979)	$\bar{Y}_A(1) = 7888$	$\bar{Y}_A(0) = 6450$



Effect of OJT on earnings

- We can compute the difference in earnings **by treatment status**:
 - we only consider the post-treatment period (1979) and compute the difference in earnings between participants and non-participants
- or we can compute the difference in earnings **over time**:
 - we only consider participants and compute the difference in their earnings between 1979 and 1975

Do we find a causal effect? **No**. We are only using two **naive estimators** that do not take potential confounding factors into account.



Naive estimators: difference in means

Difference in means: comparison of averages by treatment status. Difference between the average earnings of trainees and non-trainees in 1979 (after training):

$$DM = \bar{Y}_A^T(1) - \bar{Y}_A^C(0) = 7888 - 6450 = 1438$$

- This is not the causal effect of OJT on earnings, because individuals in T are different from individuals in C
- Non-participants are not representative of what would have happened to participants in the counterfactual situation of no training
- The workers who have the most to gain from the programme, or are most interested in self-improvement, are more likely to enrol
- We are comparing apples with oranges



Naive estimators: before-after

Before - after estimator: simply compare the earnings of participants before (1975) and after the training programme (1979):

$$BA = \bar{Y}_A^T(1) - \bar{Y}_B^T(1) = 7888 - 1512 = 6376$$

- This is not the causal effect of OJT on earnings either
- The earnings of trainees before the programme are not an approximation of their counterfactual earnings in the absence of training
- The earnings of trainees could have changed between 1975 and 1979 for other reasons, such as ageing and changes in the aggregate economic situation



Sample selection problem

- Neither naive estimator is good
- In terms of the fundamental problem of causal inference, we are not making progress
- The impact measured by the naive estimators is affected by confounding factors that we are not controlling for

Sample selection bias

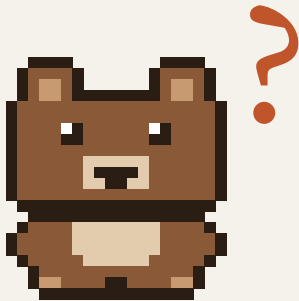


Sample selection bias

- Paradoxically, some naive comparisons of participants and non-participants find that trainees earn **less** than non-trainees
- *Sample selection bias* is the main concern: training programmes are meant for individuals with low earnings potential
- Thus, simple comparisons between participants and non-participants, not surprisingly, show lower earnings for the participants

We need new methodological approaches that allow us to **remove that bias**





Questions?

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